# Choosing and Applying a Convergence Test

Comparison, integral, ratio, and alternating-series tests

**Directions**

- Each problem names the test to use, or the structure of the terms makes it obvious. Do the computation the test asks for, then write the conclusion it licenses — and only that conclusion.
- Factorials and fixed powers point to the ratio test; a rational expression in $n$ points to limit comparison; a term you can integrate points to the integral test; alternating signs point to the alternating series test and its remainder bound.
- Give exact answers unless a decimal is asked for.

**1.** Consider $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2+3}$, to be tested by limit comparison with $\displaystyle\sum \frac{1}{n^2}$.

(a) Find $\displaystyle\lim_{n\to\infty} \frac{1/(n^2+3)}{1/n^2}$.

(b) State the conclusion the limit comparison test licenses, and explain why the limit being positive matters.

Answer: __________________

**2.** Consider $\sum_{n=1}^{\infty} ne^{-n^2}$, to be tested by the integral test with $f(x) = xe^{-x^2}$.

(a) Evaluate $\int_1^{\infty} xe^{-x^2}\, dx$ exactly.

(b) State the conditions you checked before applying the test and the conclusion you reach. Explain why the value of the integral is not the sum of the series.

Answer: ________________

**3.** Consider $\sum_{n=0}^{\infty} \frac{2^n}{n!}$.

(a) The ratio of consecutive terms simplifies to $\frac{2}{n+1}$. Find its limit as $n \to \infty$.

(b) This series has a familiar exact sum. Write it.

Answer: ________________

**4.** Consider the alternating harmonic series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$.

(a) Find $\displaystyle\lim_{n\to\infty} \frac{1}{n}$, the limit of the term sizes.

(b) Using the first 100 terms as an estimate of the sum, give the exact error bound the alternating series remainder guarantees.

(c) Justify both parts: name the two conditions the test requires and state the remainder bound you used.

Answer: ________________

**5.** Consider $\displaystyle\sum_{n=1}^{\infty} \frac{3n+1}{n^3+5}$, to be tested by limit comparison with $\displaystyle\sum \frac{1}{n^2}$.

(a) Find $\displaystyle\lim_{n\to\infty} \frac{(3n+1)/(n^3+5)}{1/n^2}$.

(b) State the conclusion, and explain how the comparison series was chosen.

Answer: ________________

**6.** Consider $\sum_{n=1}^{\infty} \frac{n!}{3^n}$. The ratio of consecutive terms simplifies to $\frac{n+1}{3}$.

(a) Find the limit of that ratio as $n \to \infty$.

(b) State the conclusion, and explain what the size of the terms themselves shows independently.

Answer: ________________

**7.** Consider $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$, to be tested by the integral test.

(a) Evaluate $\int_2^{\infty} \frac{dx}{x(\ln x)^2}$ exactly, then give a decimal to three places.

(b) Name the substitution you used, state the conditions you checked, and give the conclusion.

Answer: ________________

**8.** Consider $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$.

(a) This series has a well-known exact sum. Write it.

(b) Give the exact remainder bound guaranteed if the first five terms ($n = 0$ through $n = 4$) are used as an estimate.

Answer: ________________

**9.** Consider $\displaystyle\sum_{n=1}^{\infty} \frac{n^2\, 3^n}{n!}$.

(a) The ratio of consecutive terms simplifies to $\dfrac{3(n+1)}{n^2}$. Find its limit.

(b) Show the cancellation that produces that ratio, and state the conclusion.

Answer: ________________

**10.** Consider $\sum_{n=1}^{\infty} \frac{n+2}{n^3-n+4}$. A student begins with the ratio test and finds the limit of the ratio is 1, which settles nothing.

(a) Find $\lim_{n\to\infty} \frac{n+2}{n^3-n+4} \cdot n^2$.

(b) Write the exact sum of the comparison series $\sum_{n=1}^{\infty} \frac{1}{n^2}$.

(c) Explain why the ratio test could never have settled this series, and give the conclusion limit comparison reaches.

Answer: ____________________